

PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Software Engineering Lab 1

Requirements Engineering & UML Use-Case Modelling

SRN	PES1UG24CS381
Scenario Number	Scenario #13
Project Title	Patient Health Record Consent Management System (PHRCMS)
Primary Domain	Healthcare & Telemedicine
Target Stakeholders	Patient, Clinic Doctor, Clinic Administrator

1. Requirements Specification Table

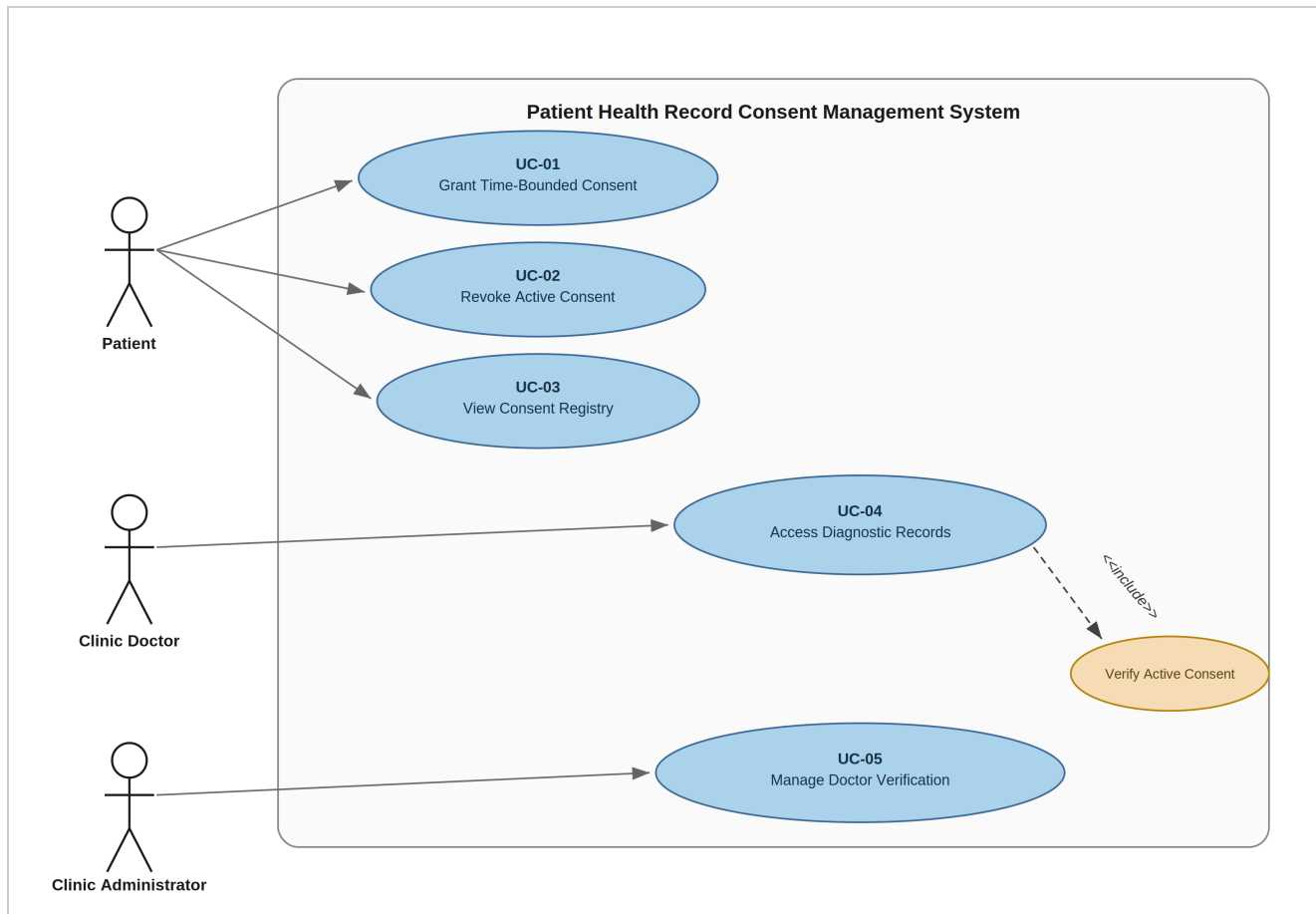
The following table specifies exactly five functional requirements (FR-001 to FR-005) and two non-functional requirements (NFR-001 to NFR-002) for the Patient Health Record Consent Management System.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a patient to grant time-bounded access permissions for specific diagnostic records to verified clinic doctors, requiring a valid expiration date and time.	High	A verified clinic doctor can view the specified diagnostic records within the designated consent time-frame, and the system automatically blocks access immediately after the consent's expiration timestamp.	Enables patients to share health data temporarily, minimizing data exposure risks.
FR-002	Functional	The system shall allow a patient to revoke any active consent permission at any time prior to its scheduled expiration.	High	Upon revocation by the patient, the status of the consent is immediately updated to 'revoked', and any subsequent attempt by the authorized doctor to access the record is blocked.	Gives patients real-time control over their data, aligning with patient-centric consent principles.
FR-003	Functional	The system shall display a real-time list of all active and expired consent permissions to the logged-in patient, including doctor details, record names, and expiration timestamps.	Medium	The consent registry page lists all consents, showing doctor name, record ID, status (active/expired/revoked), and remaining duration (or expiry date/time) accurately.	Promotes transparency, allowing patients to audit who has access to their data.
FR-004	Functional	The system shall permit a verified clinic doctor to view authorized patient diagnostic records only if there is an active, unexpired consent granted by the patient.	High	If a doctor attempts to access a record, the system grants access if an active consent exists, and denies access with an error if consent is missing, expired, or revoked.	Enforces the consent policy, preventing unauthorized data access.
FR-005	Functional	The system shall allow a Clinic Administrator to register and manage the verification status of clinic doctors in the system.	Medium	The administrator can add a doctor to the registry and toggle their status as 'verified' or 'unverified'. Unverified doctors cannot be selected by patients for consent grants.	Ensures that patients only grant access to legitimate, authorized medical professionals.
NFR-001	Nonfunctional	All consent grant, revoke, and access events shall be permanently written	High	Every consent action and record-access event	Ensures accountability,

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		to an append-only audit trail.		creates a log entry with timestamp, actor ID, and action type, which cannot be modified or deleted by any user or administrator.	data integrity, and compliance with medical privacy audits.
NFR-002	Nonfunctional	The system shall complete the verification of consent status and authorize or deny a doctor's access request in less than 500 milliseconds.	High	Under automated performance testing with 50 concurrent requests, the average response time to verify consent status and return the access decision is less than 500 milliseconds.	Enforces prompt access decision checks, ensuring doctor workflow is not delayed while maintaining strict security checks.

2. UML Use-Case Diagram

The diagram below displays the system boundary, the three target actors, and the core consent management and validation use cases, including the mandatory *Verify Active Consent* include relationship.



3. Use-Case Flow Specification: UC-01

Use Case ID	UC-01
Use Case Name	Grant Time-Bounded Consent
Primary Actor	Patient
Purpose	Allows a patient to select specific diagnostic records and grant time-bounded access to a verified clinic doctor, with access automatically revoking after the designated expiration timestamp.

Preconditions

- The Patient is authenticated and logged into the Patient Health Record Consent Management System.
- The Clinic Doctor is registered in the system and verified by the Clinic Administrator.
- The Patient has one or more diagnostic records linked to their account.

Main Success Scenario

1. Patient selects the 'Grant Record Access' option from the dashboard.
2. System displays a form requesting selection of a verified doctor, selection of diagnostic records, and a specific expiration date and time.
3. Patient searches the verified registry and selects a verified Clinic Doctor.
4. Patient selects one or more specific diagnostic records (e.g., 'Blood Lab Report - 2026-08-20') from their medical history.
5. Patient enters a valid future date and time for the consent's expiration.
6. Patient submits the consent grant request.
7. System validates that the selected doctor is verified, the records belong to the patient, and the expiration timestamp is in the future.
8. System creates the active consent record with the selected parameters and calculated expiration timestamp.
9. System permanently writes the consent grant event (timestamp, Patient ID, Doctor ID, Record IDs, Expiration Timestamp, Action = 'GRANT') to the append-only audit trail.
10. System displays a success message confirming that access has been granted to the doctor until the specified expiration date and time.

Alternate Flows

3a. Doctor not verified:

- 3a1. System checks the doctor's status and determines they are unverified.
- 3a2. System displays an error: 'The selected doctor is currently unverified. Access can only be granted to verified clinic doctors.'
- 3a3. System prompts the patient to select a different doctor.
- 3a4. Flow returns to step 3 of the Main Success Scenario.

5a. Expiration time in the past or invalid:

- 5a1. System checks the entered expiration timestamp and determines it is in the past or formatted incorrectly.

- 5a2. System displays an error: 'Expiration timestamp must be a valid date and time in the future.'
- 5a3. System prompts the patient to re-enter the expiration date and time.
- 5a4. Flow returns to step 5 of the Main Success Scenario.

Postconditions

- A consent record is stored in the system database containing the Patient ID, Doctor ID, Record IDs, Grant Timestamp, Expiration Timestamp, and Status set to 'Active'.
- A permanent, unmodifiable event entry is written to the append-only audit trail.
- The designated Clinic Doctor is authorized to access the specified diagnostic records until the expiration timestamp is reached.